

5(a) On Earth's surface,

$$v = r \omega$$

$$T = 24 \text{ hr}$$

$$v = r \omega = (6.4 \times 10^6) \left(\frac{2\pi}{T} \right) = (6.4 \times 10^6) \left(\frac{2\pi}{24(3600)} \right) = 465.42 \text{ m s}^{-1}$$

M1

$$\text{Kinetic energy} = \frac{1}{2} m v^2 = \frac{1}{2} (100) (465.42)^2 = 1.0831 \times 10^7 \text{ J}$$

Kinetic energy at low Earth orbit

$$= 1.0831 \times 10^7 + 1.5 \times 10^9 - m \Delta \phi \square \square$$

$$= 1.0831 \times 10^7 + 1.5 \times 10^9 - 100 [6.67 \times 10^{-11} (6.0 \times 10^{24}) \left(-\frac{1}{(1.5 + 6.4) \times 10^6} + \frac{1}{6.4 \times 10^6} \right)]$$

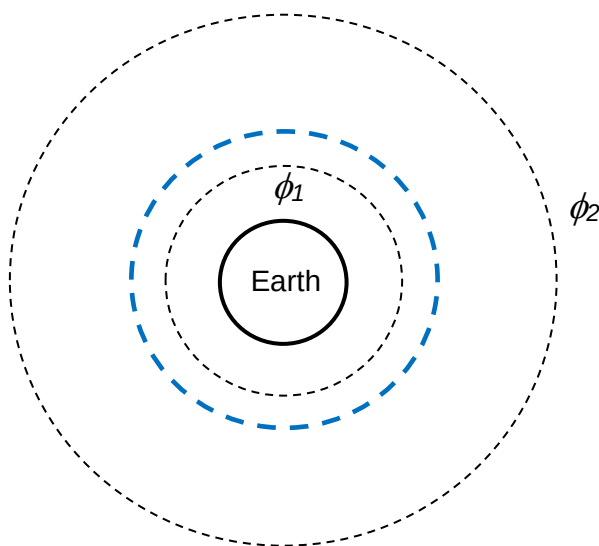
M1

$$= 3.2 \times 10^8 \text{ J}$$

A1

B1

(b)



Equipotential line nearer to ϕ_1

Concentric circle

(c) The negative of the gravitational force on nanosatellite by Earth.

B1

[Turn over]

(d)

If the body has sufficient energy to escape to infinity, the kinetic energy must be equal or greater than the magnitude of the gravitational potential energy. The total energy can thus be equal to or more than zero, which is represented by point D or E.

B1

B1